



Masonry in Construction

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Introduction to Masonry

- **Masonry** is the construction of walls, partitions, foundations and other structural elements using individual units such as **bricks, stones, concrete blocks or other masonry units**.
- The units are generally bonded together using **mortar**.
- Masonry provides:
 - ❖ Strength and stability
 - ❖ Protection from weather
 - ❖ Fire resistance
 - ❖ Thermal and sound insulation
 - ❖ Architectural appearance



CLASSIFICATION OF MASONRY — BASED ON MATERIAL —



1. BRICK MASONRY



- Uses bricks as masonry units
- Commonly used for walls and foundations



2. STONE MASONRY



- Uses natural stones
- Suitable for heavy and durable construction



3. CONCRETE BLOCK MASONRY



- Uses precast concrete blocks
- Faster construction and uniform dimensions



4. AAC BLOCK MASONRY



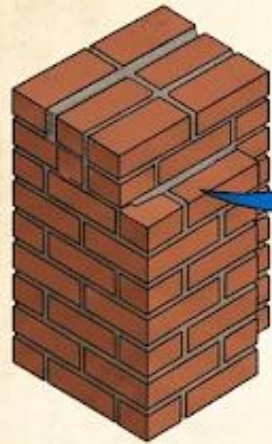
- Uses Autoclaved Aerated Concrete blocks
- Lightweight and provides good thermal insulation





Terms used in Masonry

1. FACE

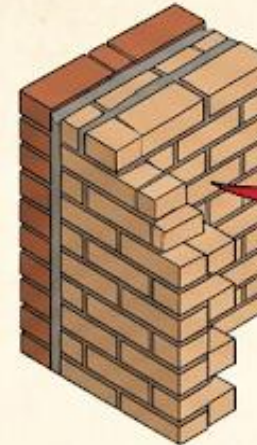


SURFACE OF THE WALL EXPOSED TO THE WEATHER.



MATERIAL USED IN THE FACE OF THE WALL IS KNOWN AS FACING.

2. BACK

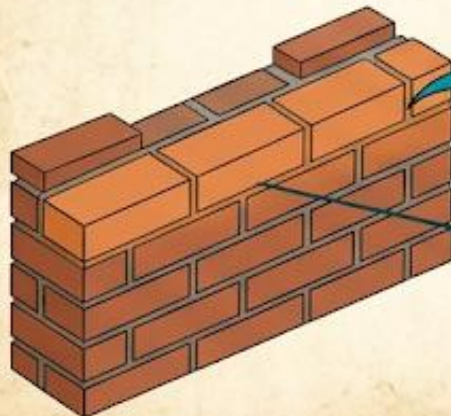


SURFACE OF THE WALL WHICH IS NOT EXPOSED TO THE WEATHER.

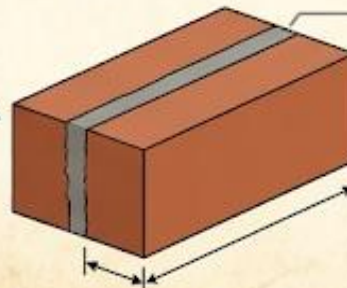


MATERIAL USED IN THE BACK OF THE WALL IS KNOWN AS BACKING.

3. COURSE



HORIZONTAL LAYER OF MASONRY UNIT.



THICKNESS OF ONE MORTAR JOINT.
THICKNESS OF MODULAR BRICK.

THICKNESS OF THICKNESS OF MODULAR BRICK
A COURSE = +
THICKNESS OF MODULAR BRICK
+ THICKNESS OF ONE MORTAR JOINT
(IN BRICK MASONRY)



Terms used in Masonry

Corbel

- ❖ Projecting stone to serve as support for truss or weather shed etc.
- ❖ Should extend at least two-thirds of their length.
- ❖ Generally moulded and given ornamental treatment.



Cornice

- ❖ An ornamental moulding near the top of a wall just below the ceiling.
- ❖ Penetrates the full width of the wall; weathered and throated to dispose of rainwater.



Coping

- ❖ Covering of stone, brick or terracotta placed on exposed top of parapet or compound wall to prevent seepage of water.
- ❖ It is weathered and throated.





Terms used in Masonry

Weathering

Provision of slope on the upper surface as sills, cornices.

Throating

Groove provided on the underside of projecting elements such as sills, cornices so that rain water can be discharged clear off the wall surface.

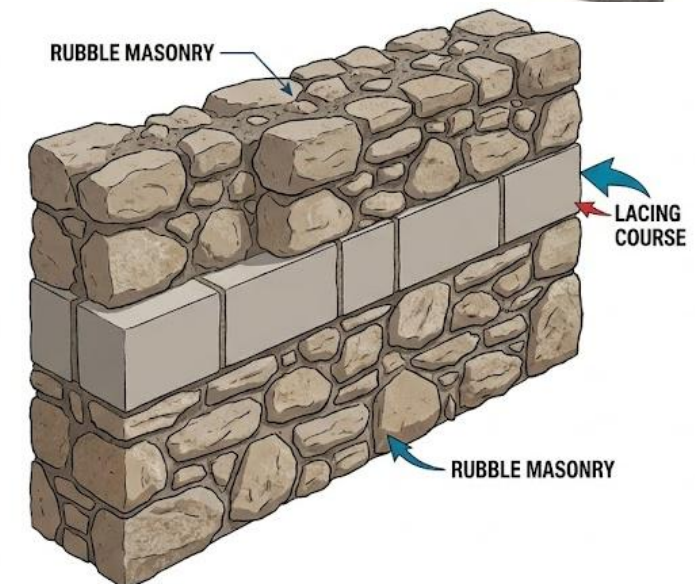
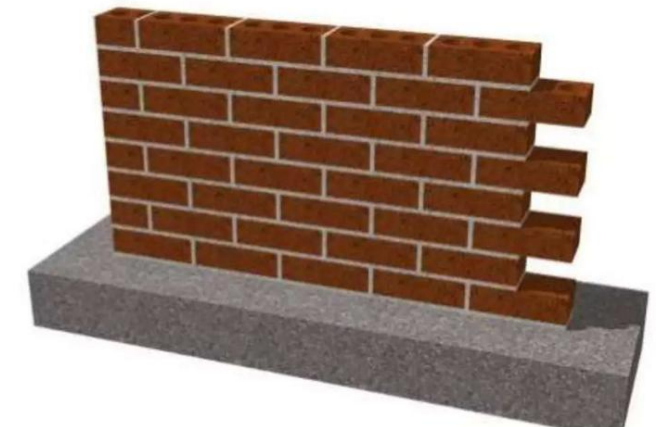
Toothing

Bricks left projecting in alternate course for purpose of bonding future masonry work.

Lacing Course

Horizontal course of stone blocks provided to strengthen a wall made of irregular courses of small stones.

Toothing





Stone Masonry

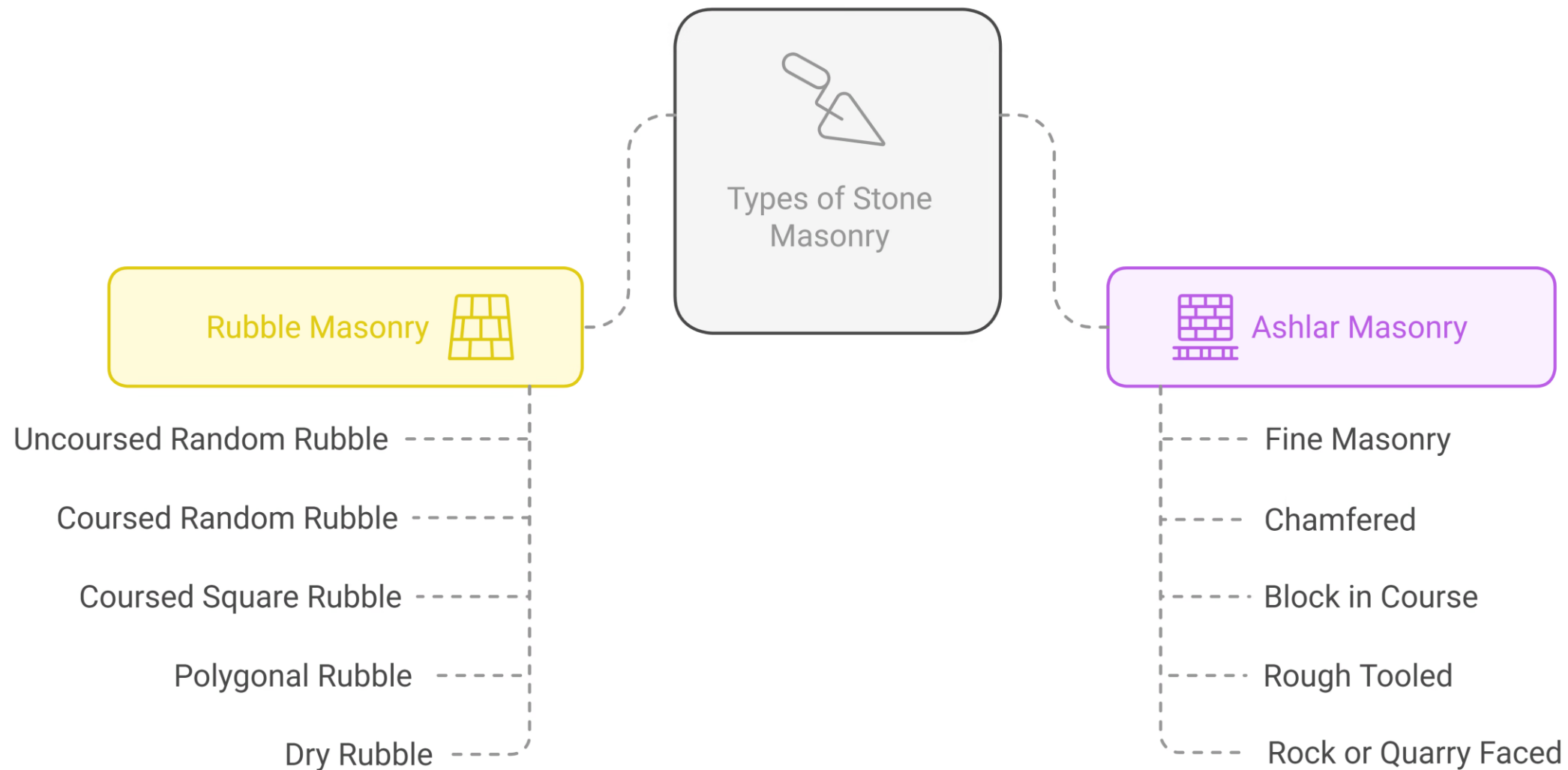
CLASSIFICATION OF STONE MASONRY

- ❖ **Rubble masonry** Wall is made up of stones of irregular sizes and shapes i.e. stones are roughly dressed. The stones from the quarry are broken into small pieces and are directly used in construction work.
- ❖ **Ashlar masonry** Wall is made of accurately dressed stones with extremely fine bed and end joints. Block may be either square and rectangular shaped.





Stone Masonry





Uncoursed Random Rubble Masonry

- It is a type of stone masonry in which **rough and irregularly shaped stones** are used without arranging them in regular horizontal courses.
- The stones are generally **undressed or only slightly dressed**.
- Stones of different **sizes and shapes** are used.
- The stones are arranged carefully so that they fit closely with one another.
- The gaps between the stones are filled with **mortar and small pieces of stone**.
- Larger stones are generally placed at the **bottom and corners** of the wall.





Coursed Random Rubble Masonry

- ❖ Stones are laid in regular horizontal layers or courses.
- ❖ Each course consists of stones of approximately similar height.
- ❖ The height of each course generally ranges from 300 to 900 mm.
- ❖ Stones are roughly dressed to fit neatly into rows.
- ❖ Stones may be obtained from local quarries, such as granite, limestone and sandstone.
- ❖ Joints are generally 10–20 mm thick.
- ❖ It provides a strong, neat and uniform appearance.
- ❖ It is more organized and attractive than uncoursed rubble masonry.





Coursed Square Rubble Masonry

- Stones are laid in regular horizontal courses.
- The edges of the stones are made nearly straight, but they are not perfectly square.
- Stones have a more uniform and structured appearance than ordinary coursed rubble masonry.
- Joints between stones are generally 10–15 mm thick.
- It provides a neater and more refined finish while retaining the natural appearance of stone.
- It is stronger and more attractive than uncoursed rubble masonry.





Polygonal Rubble Masonry

- ❖ Stones are roughly hammer-pitched into irregular polygonal shapes and are generally without prominent stratification.
- ❖ The irregular faces of the stones are exposed to create irregular joints in different directions, giving an aesthetic appearance.
- ❖ Long continuous joints are avoided, and joints are broken as much as possible to prevent vertical alignment.
- ❖ Joints of adjacent courses should not lie one above another, as this considerably reduces the strength of the masonry.





Dry Rubble Masonry

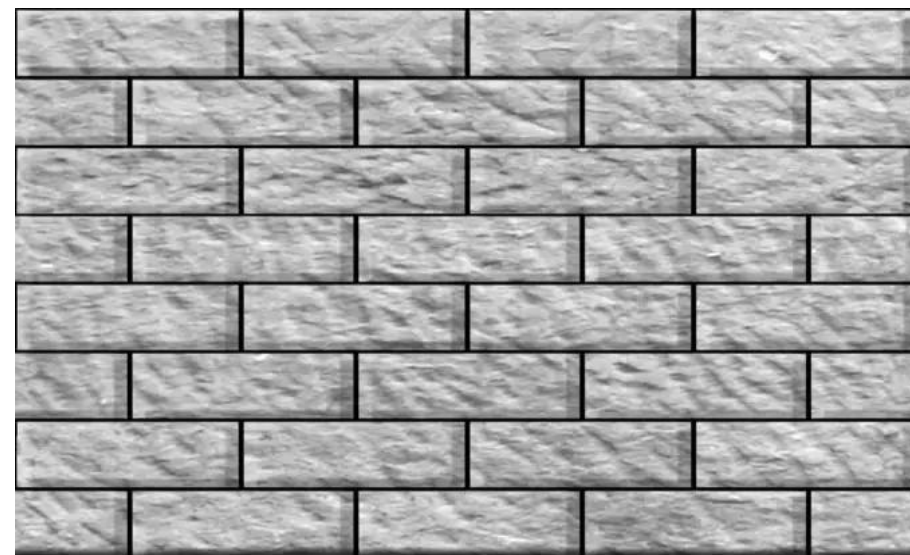
- ❖ It is constructed like random rubble masonry without mortar, with stones tightly packed together.
- ❖ Small stone pieces are used to fill the gaps and hollow spaces between larger stones.
- ❖ It is economical because no mortar is required, but skilled labour is needed for proper packing of stones.
- ❖ The height of stone masonry should not exceed 6 m; if it exceeds 3 m, three adjacent courses are provided at 3 m intervals in square rubble masonry.
- ❖ It is mainly used for non-load-bearing walls, retaining walls, and pitching of earthen dams and canal slopes.





Ashlar Fine Masonry

- ❖ Stones are carefully dressed into regular rectangular or square blocks with uniform size and shape.
- ❖ The exposed faces are finely dressed and smooth, giving the masonry a neat and attractive appearance.
- ❖ The stones are laid in regular courses, with very thin and uniform mortar joints.
- ❖ The beds and sides of each stone are accurately dressed to ensure close contact and proper bonding.
- ❖ It is used for important buildings, architectural works, facades, columns and ornamental structures where a high-quality finish is required.





Ashlar Rough Tooled Masonry

- ❖ Stones are accurately dressed into regular rectangular or square blocks with uniform beds and sides.
- ❖ The exposed face of the stone is roughly tooled, leaving a series of small marks or grooves on the surface.
- ❖ The edges, corners and margins of the stones are dressed neatly to maintain a regular appearance.
- ❖ Stones are laid in regular horizontal courses with thin and uniform mortar joints.
- ❖ It is commonly used for building facades, boundary walls, columns and architectural works where a neat but less polished finish is required.





Ashlar Chamfered Masonry

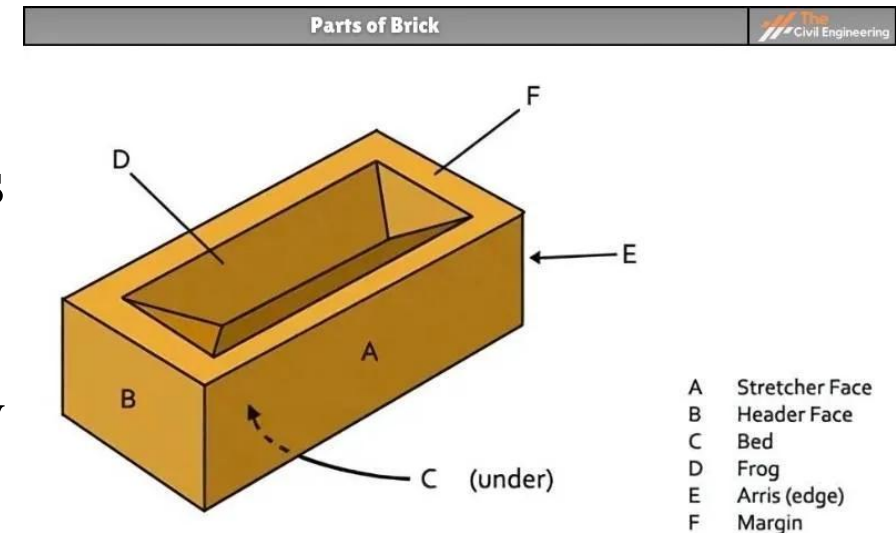
- ❖ Stones are dressed into regular rectangular or square blocks with accurately finished beds and sides.
- ❖ The exposed edges of each stone are beveled or chamfered, forming a sloping surface around the face.
- ❖ The central portion of the exposed face is kept flat or slightly rough, creating a decorative appearance.
- ❖ Stones are laid in regular horizontal courses with thin and uniform mortar joints.
- ❖ It is mainly used for architectural facades, public buildings, monuments, and decorative masonry work.





Brick Masonry

- ❖ **Header:** A full stone or brick unit whose length is perpendicular to the face of the wall. In case of stone masonry, header is sometimes called Through stone.
- ❖ **Stretcher:** A full stone or brick unit whose length is parallel to the face of the wall.
- ❖ **Header Course:** A course of brick showing only headers on the exposed face of the wall.
- ❖ **Stretcher Course:** A course of brick showing only stretchers on the exposed face of the wall.





Brick Masonry

Bond

- Overlapping of bricks or stones in alternate courses, so that no continuous vertical joints are formed and individual units are tied together.

Quoins

- Exterior angle or corner of a wall.
- The stones or bricks forming the quoins are known as stone quoins or quoin bricks.
- If the quoin's width is parallel to the face of the wall, it is known as a quoin header.
- If the length is parallel to the face of the wall, it is called a quoin stretcher.

Queen Closer

- Portion of brick obtained by cutting a brick length-wise into two portions.
- It is half as wide as the full brick.



Brick Masonry

King Closer: If a brick is cut in such a way that the width of one end becomes half that of a full brick, while the width at the other end is equal to the full width, then it is called as *king closer*. It is obtained by cutting out a triangular portion of the brick between the centre of one end (width side) and the centre of the other end (lay side). Thus, it has half header and half stretcher face.

Beveled Closer: It is a form of king closer in which the whole length of the brick (i.e. stretcher face) is chamfered or beveled in such a way that half width is maintained at one end and full width is maintained at the other end.

Mitred Closer: It is a portion of a brick whose one end is cut splayed or mitred for full width. The angle of splay may vary from 45° to 60°. Thus, one longer face of the mitred closer is of full length of the brick while the other longer face is smaller in length.



Brick Masonry

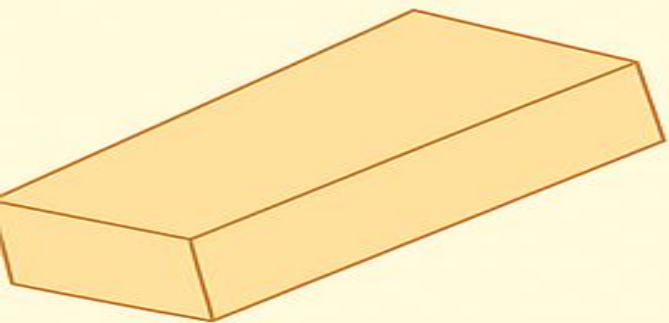
Bat: When a brick is cut across the width, the resulting piece is called *bat*. Thus, a bat is smaller in length than the full brick.

Half Bat: If the length of the bat is equal to half the length of the original brick, it is known as *half bat*.

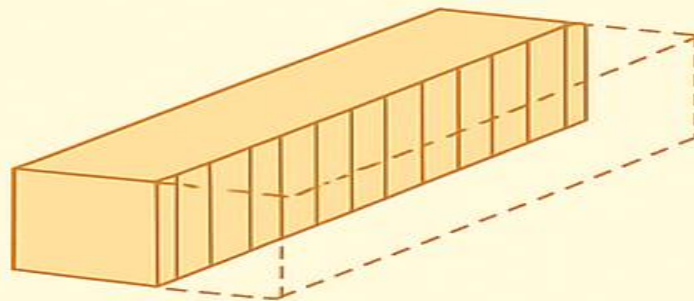
Three Quarter Bat: It is a form of brick bat having its length equal to three quarter of the length of a full brick.

Beveled Bat: A brick bat is called *beveled bat*, when its width has beveled.

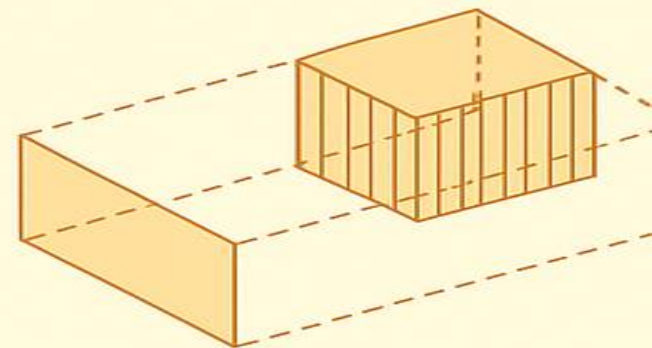
Perpend: The vertical joint between two adjacent bricks or blocks in the same horizontal course.



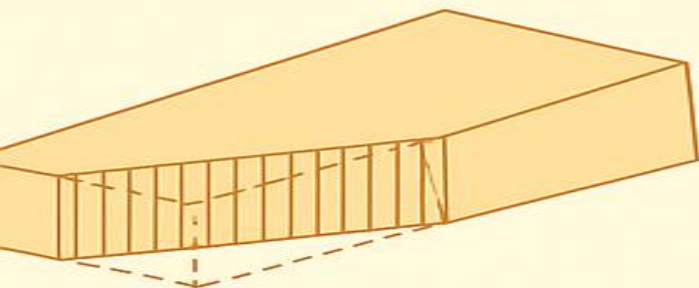
(a) Full Brick



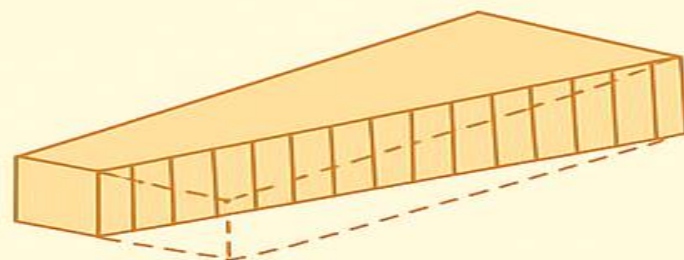
(b) Queen-Closer (Half)



(c) Queen-Closer (Quarter)



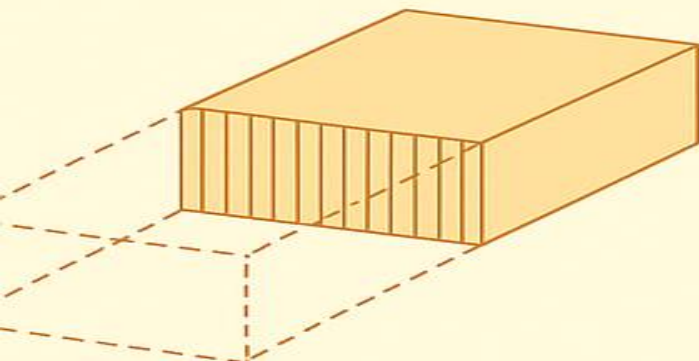
(d) King Closer



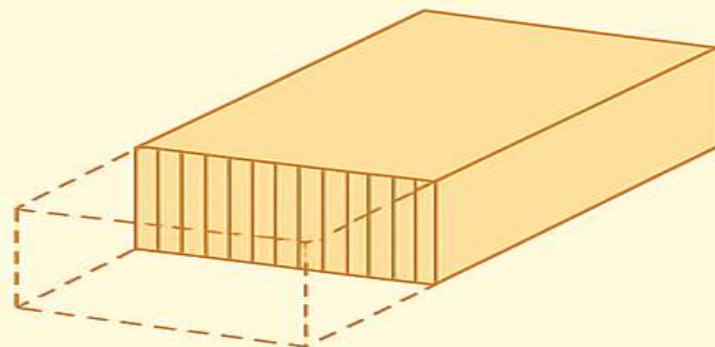
(e) Bevelled Closer



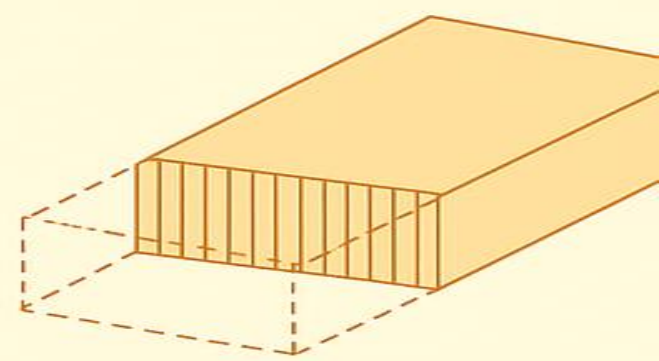
(f) Mitred Closer



(g) Half Bat

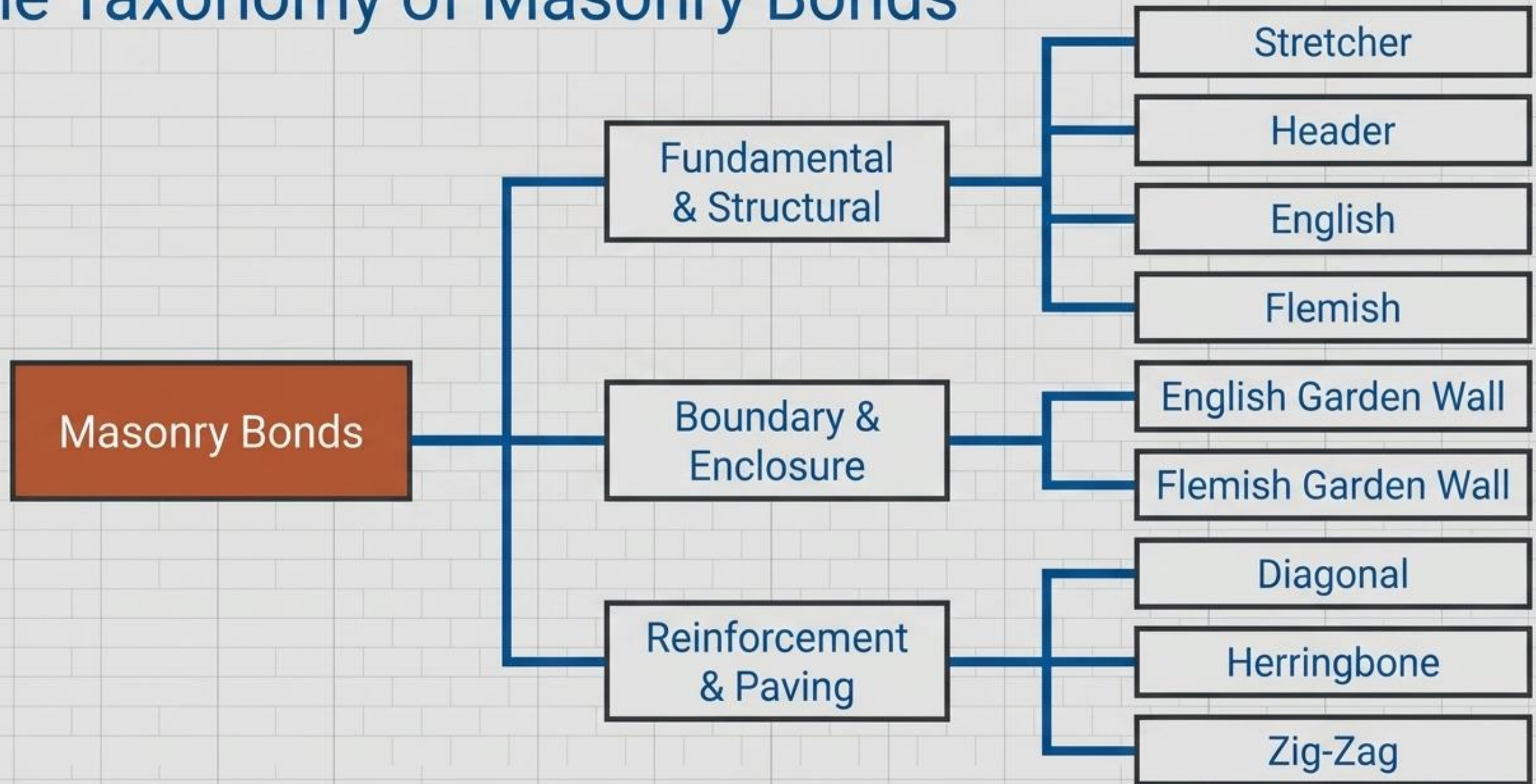


(h) Three Quarter Bat



(i) Bevelled Bat

The Taxonomy of Masonry Bonds

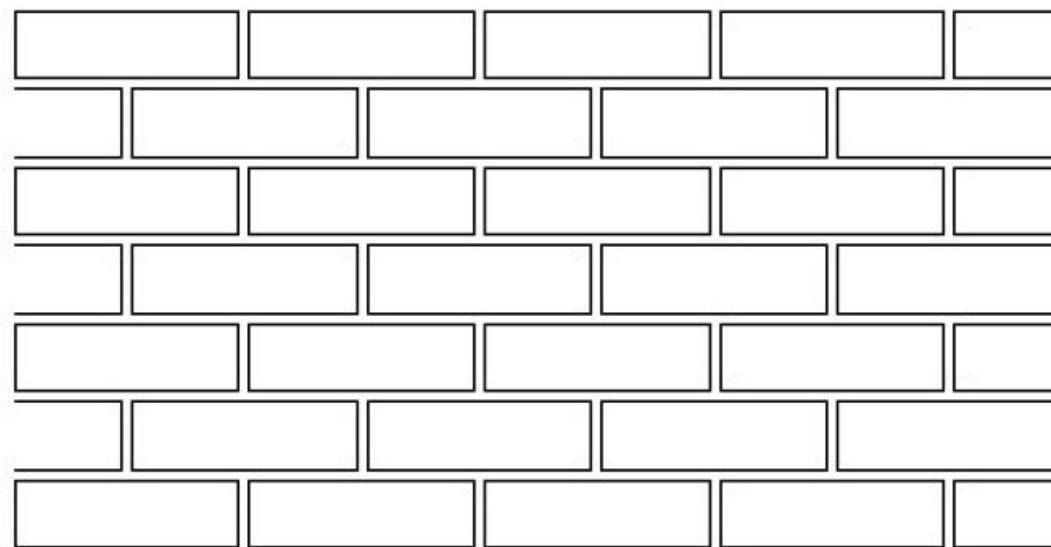




Bonds in Brick Masonry

Stretcher Bond:

- In this bond, only the stretchers are visible on the face of the wall.
- The bricks are laid with their longer faces parallel to the wall face.
- Each brick in one course is placed with its joint midway over the brick below, providing proper bonding.
- It is suitable mainly for half-brick-thick partition walls and cavity walls.
- It is not suitable for load-bearing walls because there is no proper transverse bonding between the two faces.

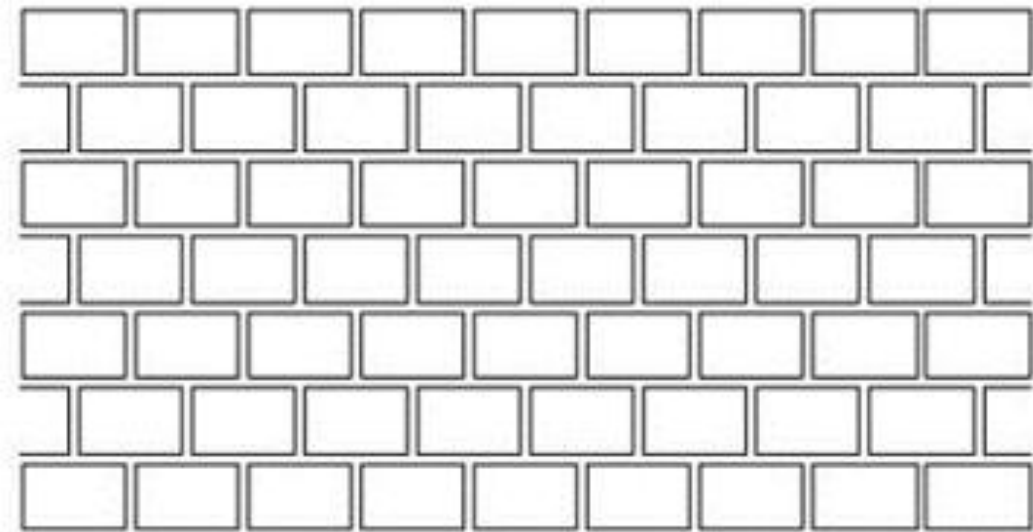




Bonds in Brick Masonry

Header Bond:

- ❖ All bricks are laid as headers, with their short ends visible on the wall face.
- ❖ Bricks are placed perpendicular to the wall face.
- ❖ It provides good bonding and transverse strength.
- ❖ Mainly used for thick walls and foundations.
- ❖ Queen closers may be used at corners to maintain the bond.
- ❖ It is not suitable for thin walls.

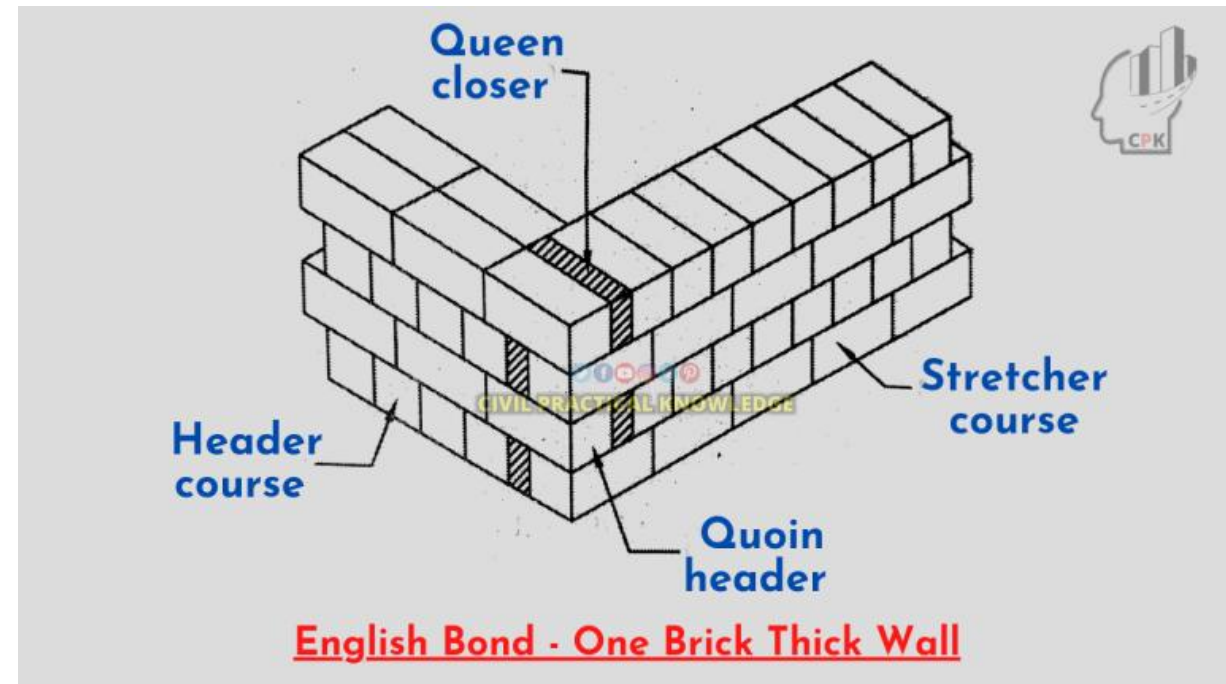




Bonds in Brick Masonry

English Bond:

- ❖ Consists of alternate courses of headers and stretchers.
- ❖ Headers and stretchers are arranged in separate horizontal courses.
- ❖ Queen closers are placed next to the quoin header to maintain proper bonding.
- ❖ It provides high strength and stability.
- ❖ It is commonly used for load-bearing and thick walls.
- ❖ It is considered one of the strongest brick masonry bonds.

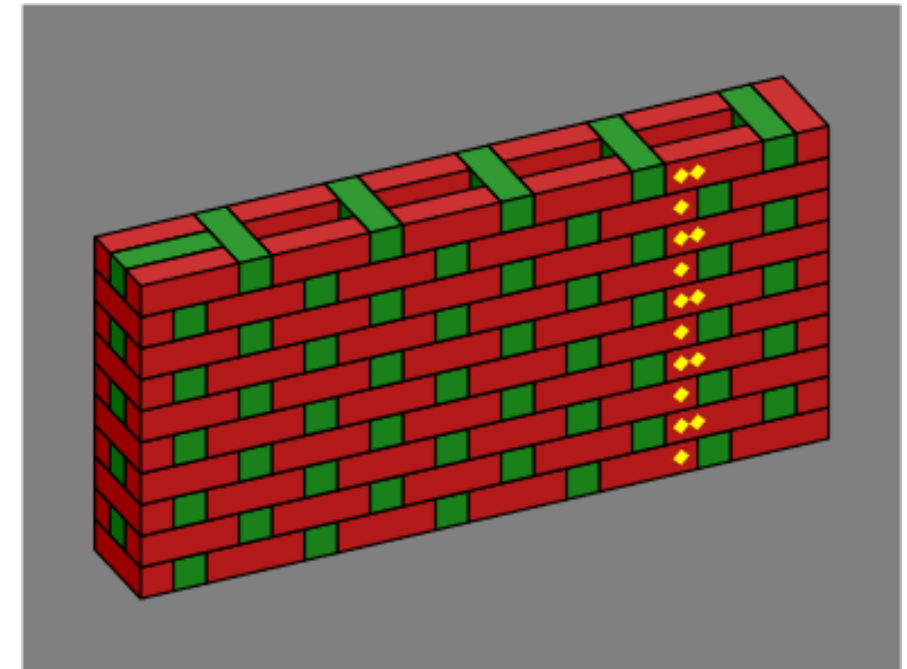




Bonds in Brick Masonry

Flemish Bond:

- ❖ In Flemish bond, headers and stretchers are arranged alternately in every course.
- ❖ Each header is placed at the centre of the stretcher in the course below, creating a staggered arrangement of vertical joints.
- ❖ Queen closers are generally provided next to the quoin headers to maintain proper bonding at the corners.
- ❖ It provides a good appearance on both faces of the wall and is commonly used where appearance is important.
- ❖ It is more difficult to construct than English bond and generally has slightly lower strength due to the arrangement of bricks.



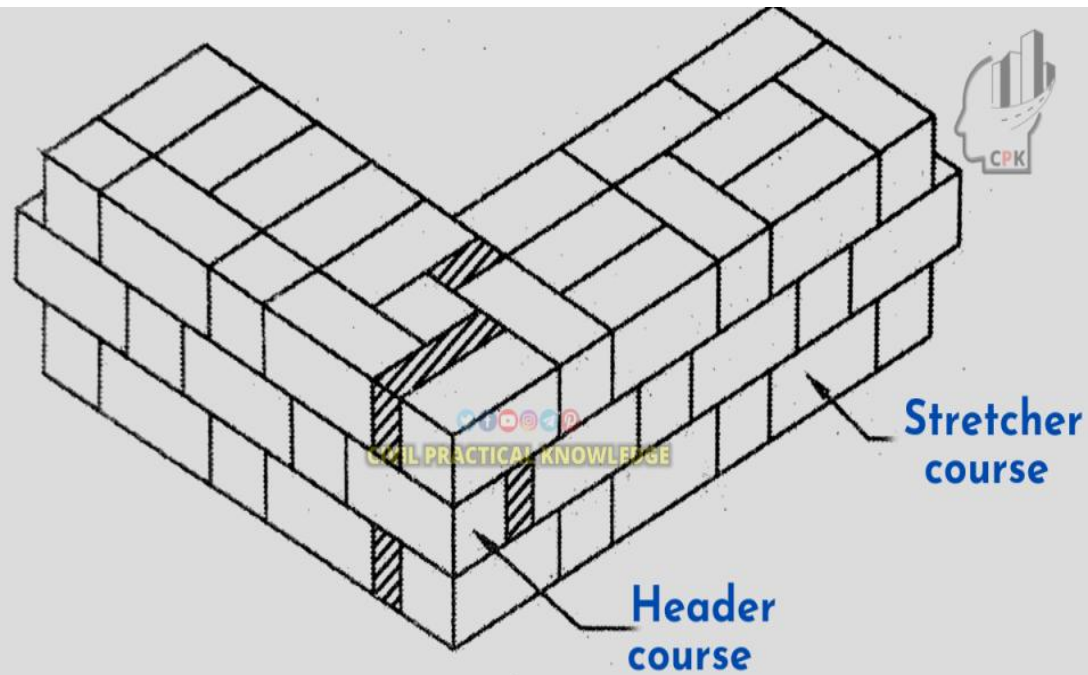


Bonds in Brick Masonry

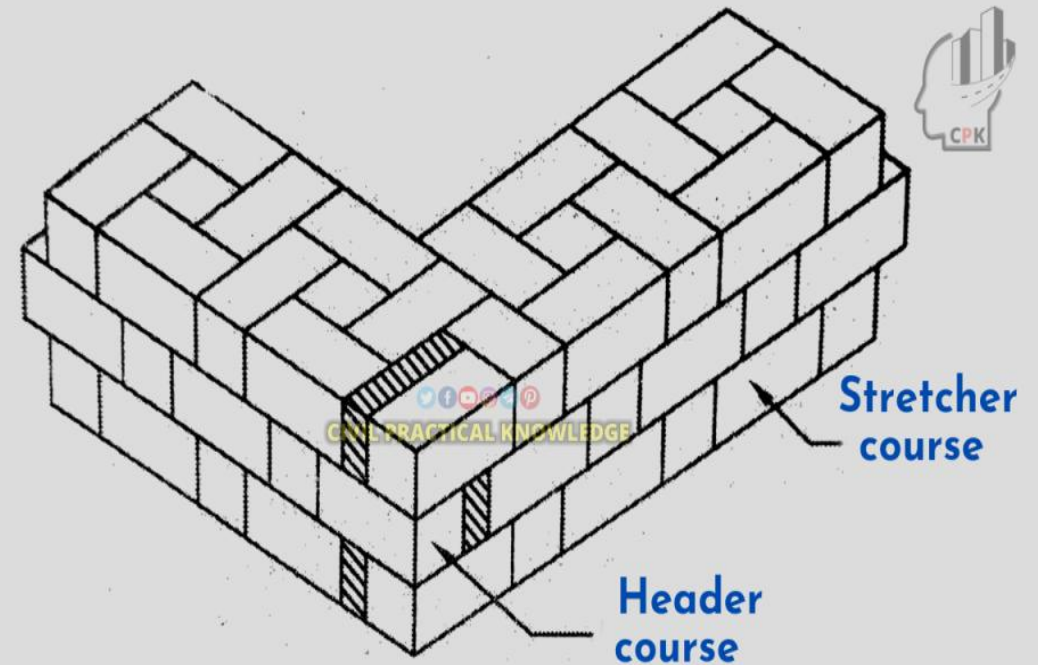
Basis	Single Flemish Bond	Double Flemish Bond
Arrangement	Flemish bond on the front face, while the rear portion is generally laid in English bond.	Flemish bond is provided on both the front and rear faces of the wall.
Appearance	Flemish appearance is visible mainly on one face.	Flemish appearance is visible on both faces.
Strength	Stronger than double Flemish bond because the backing provides better bonding.	Generally weaker than single Flemish bond for the same wall thickness.
Construction	Easier to construct than double Flemish bond.	More difficult because headers and stretchers must be properly arranged on both faces.
Use	Used where good appearance on one face and higher strength are required.	Used where good appearance on both faces is required.



Bonds in Brick Masonry



Single Flemish Bond - One and a half Brick Thick Wall



Double Flemish Bond - One and a half Brick Thick Wall



Bonds in Brick Masonry

Garden Wall Bond

- This type of bond is suitably adopted for one brick thick wall which may act as a garden wall or a boundary wall. In garden wall bond, it is possible to build uniform faces for a wall without much labour or expense. This type of bond is not so strong as English bond and its use is restricted to the construction of dwarf walls or other similar types of walls which are not subjected to large stresses. On accounts of its good appearance, this bond is sometimes used for the construction of the outer leaves of cavity walls.
- There are two types of **garden wall bond**:
 - (a) English garden wall bond
 - (b) Flemish garden wall bond



Bonds in Brick Masonry

English Garden Wall Bond

- The general arrangement of bricks in this type of bonding is similar to that of English bond, except that the heading courses are inserted only at every fourth or sixth course.
- Usually, the arrangement consists of one course of headers to three courses of stretchers.
- A queen closer is placed next to the quoin header of the heading course to provide the necessary lap.

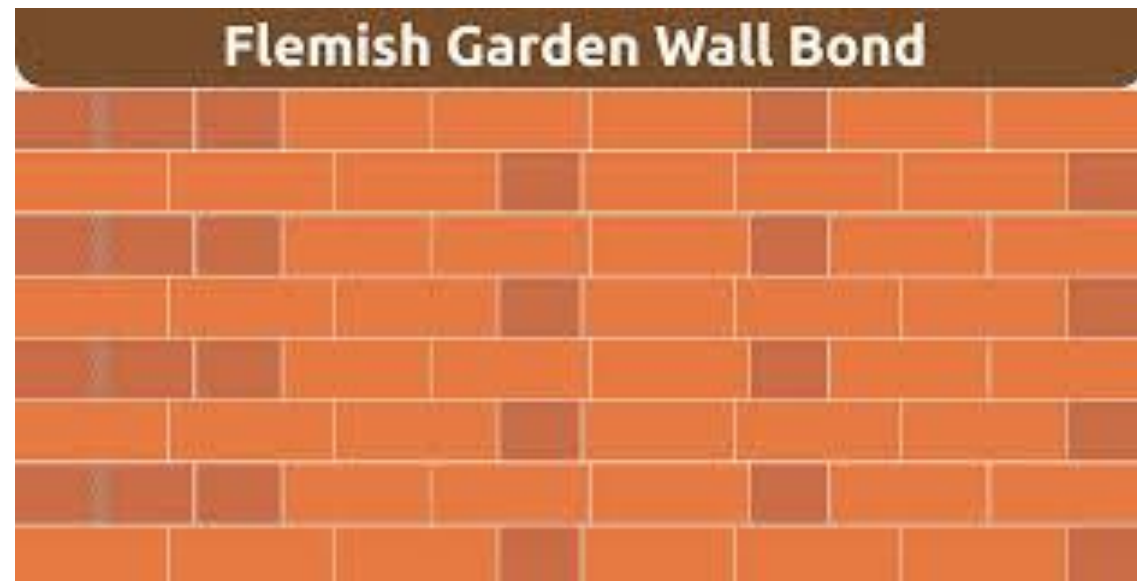




Bonds in Brick Masonry

Flemish Garden Wall Bond

- This consists of alternate course composed of one header to three or sometimes even five stretchers in series throughout the length of the courses.
- Each alternate course contains a three quarter bat placed next to the quoin header and a header is laid over the middle of each central stretcher.





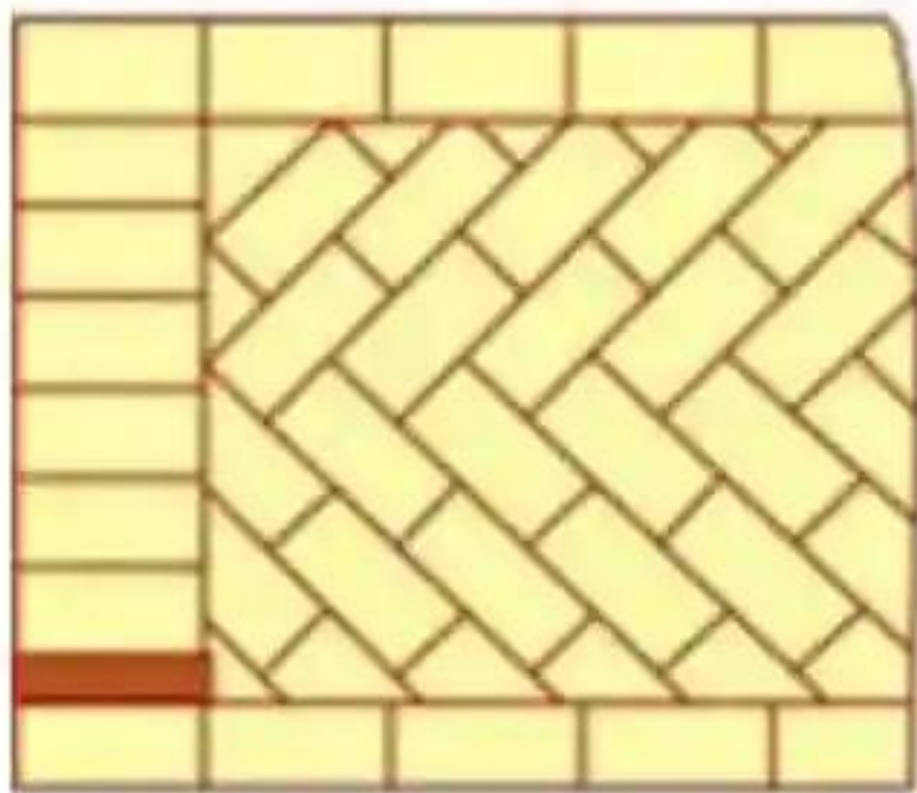
Bonds in Brick Masonry

Raking bond:

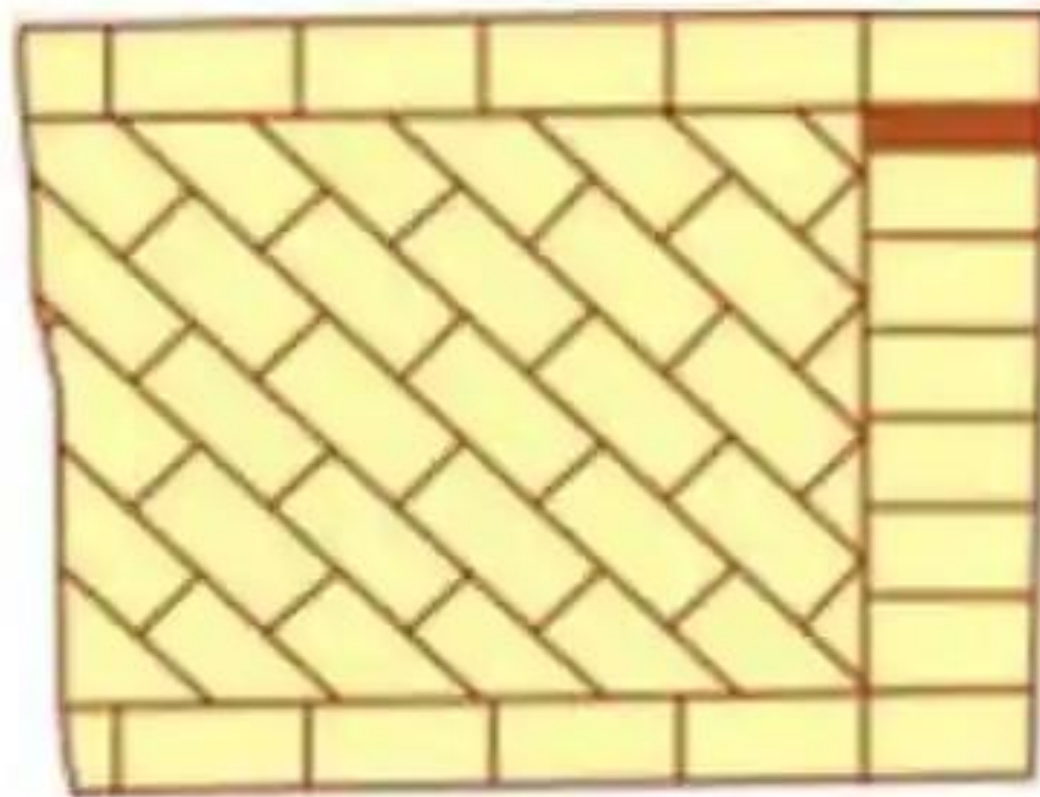
This is a bond in brick work in which the bonding bricks are laid at any angle other than zero or ninety degrees. This arrangement helps to increase the longitudinal stability of thick walls built in English bond. In this arrangement of bonding, the space between the external stretchers of a wall is filled with bricks inclined to the face of the wall. This bond is introduced at certain intervals along the height of a wall.

There are two common forms of raking bond:

- ✓ Herring bone bond
- ✓ Diagonal bond



Herring Bone Bond



Diagonal Bond

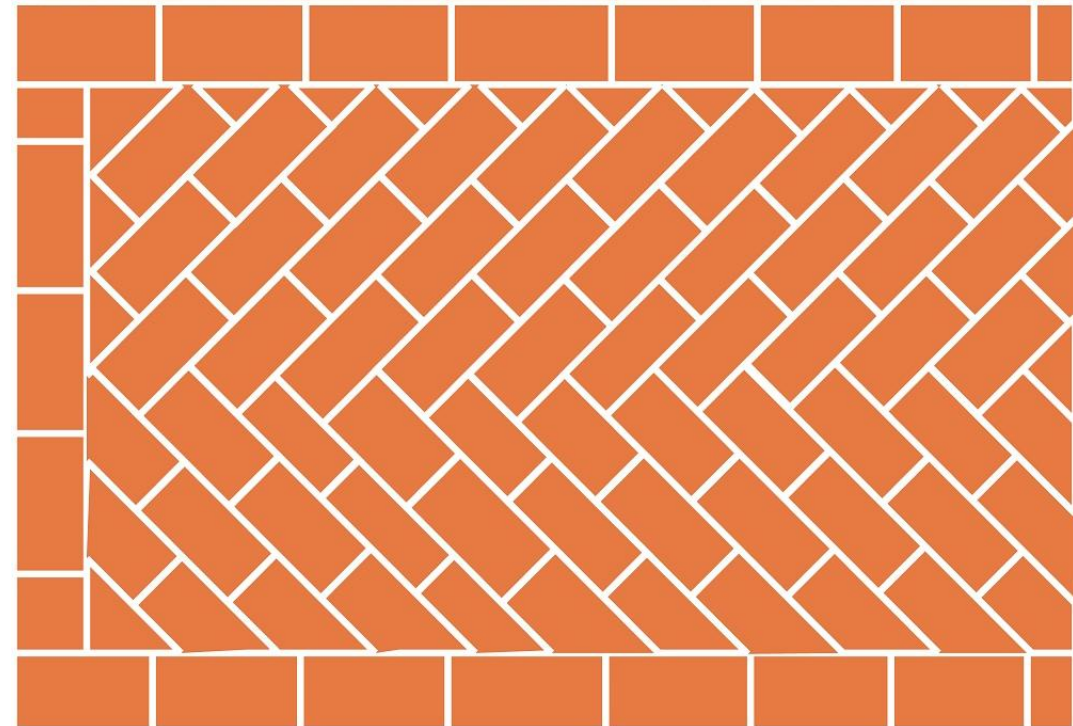
Raking Bond



Bonds in Brick Masonry

Herringbone bond

- ❖ A brickwork bond in which the exposed brickwork is bonded to the heart of the wall by concealed courses of bricks laid diagonally to the faces of the wall in a herringbone pattern.
- ❖ In this arrangement of brickwork, bricks are laid in courses inclined at 45° in two directions from the centre. This bond is also commonly used for brick pavings.

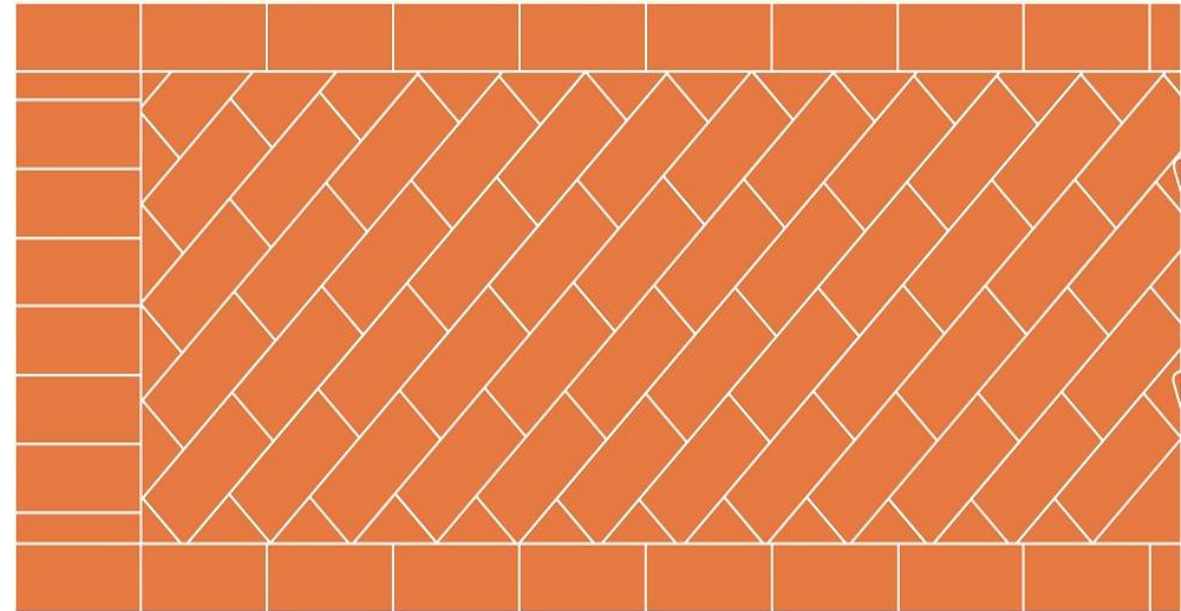




Bonds in Brick Masonry

Diagonal bond.

- ❖ This bond is best suited for walls which are 2 to 4 brick thick. This bond is usually introduced at every fifth or seventh course along the height of the wall.
- ❖ In this bond, the bricks are placed end to end in such a way that extreme corners of the series remain in contact with the stretchers.





Bonds in Brick Masonry

Zig-Zag bond:

- ❖ This is similar to herringbone bond with the only difference that in this case the bricks are laid in a zig-zag fashion.
- ❖ This is commonly adopted in brick-paved flooring.





Bonds in Brick Masonry

Basis	English Bond	Flemish Bond
Arrangement	Alternate courses of headers and stretchers are provided.	Headers and stretchers are alternately placed in the same course.
Strength	Stronger and more stable.	Comparatively weaker than English bond.
Appearance	Simple and less attractive.	More attractive and decorative.
Construction	Easier to construct.	More difficult to construct accurately.
Skill required	Requires comparatively less skill.	Requires more skilled workmanship.
Number of bricks	Requires slightly more bricks for the same wall area.	Generally requires fewer bricks.
Vertical joints	Vertical joints are properly broken between courses.	Vertical joints are more complex and require careful bonding.
Cost	Generally economical due to easier construction.	Comparatively costly due to skilled labour and careful laying.
Strength of wall	Suitable for thicker and heavily loaded walls.	Suitable mainly for moderate loads and decorative masonry.

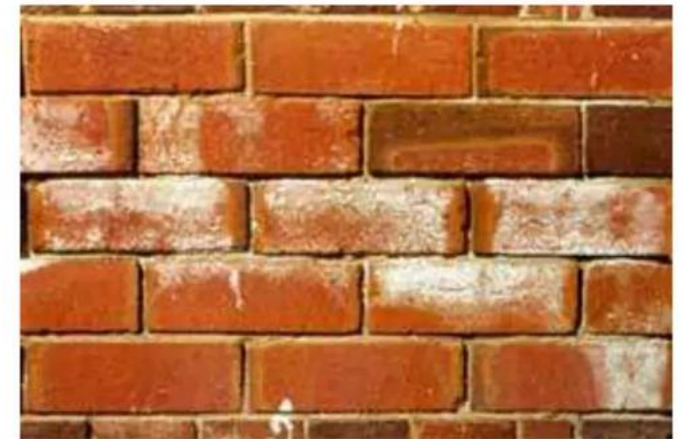


Bonds in Brick Masonry

Sulphate attack: Sulphate salts present in brick react with hydraulic lime in case of lime mortar and with alumina present in cement mortar, increasing the volume of mortar and causing chipping and spalling of bricks.



Crystallization of salts: If bricks containing soluble salts get dissolved with water, they appear in the form of fine white crystals on the surface; this is called efflorescence.



Drying shrinkage: Drying shrinkage may lead to the formation of cracks.

THANK
YOU